

The Local Occurrence Hypothesis

A Preregistered SAT-Specific Test Protocol after Gate Challenge Cycle I

MAAT Structural Selection Series – Paper 72

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Abstract

Gate Challenge Cycle I has been completed and archived with DOI <https://doi.org/10.5281/zenodo.21062386>. Paper 69 executed the frozen MAAT v1.7 gate on a public SATLIB smoke subset and did not support the gate hypothesis in SAT: the gate did not beat score-with-R with a positive confidence lower bound and lost clearly to MOMS. Paper 70 diagnosed the result as missing decision-local structural information. Paper 71 then showed that most local SAT channels are reconstructible from the existing support basis $(H, B, S, V, R_{\text{rob}}, G_{\text{gate}})$, leaving only weak occurrence/degree residuals as hypothesis-generating candidates.

The present paper does not report new solver results. It does not modify Paper 68, does not rescue Paper 69, does not introduce MAAT v1.8, and does not retune the gate. It is a preregistration protocol for the next SAT-specific test. The frozen hypothesis is that decision-local occurrence/degree residuals carry additional SAT-specific decision information beyond the frozen v1.7 gate. The null hypothesis is that these residual channels provide no robust improvement over frozen gate, score-with-R, or classical SAT heuristics.

The protocol freezes the local channel definitions L_{occ} and L_{deg} , their residualization against $(H, B, S, V, R_{\text{rob}}, G_{\text{gate}})$, the combined channel $L_{\star} = \frac{1}{2}[z(D_{\text{res}}) - z(O_{\text{res}})]$, the Gate+ L combination rule, SATLIB/DIMACS dataset rules, family-balanced splits, conflict/runtime budgets, baselines, bootstrap confidence intervals, the shuffled-local-channel null, and explicit success and failure criteria. No Gate+ L solver run is executed in this paper. Execution is reserved for a future Paper 73 after this protocol has been frozen as a public release.

1 Purpose and Guardrail

Paper 72 is a preregistration paper. It belongs to the methodological sequence:

external test $\rightarrow$ diagnosis $\rightarrow$ hypothesis narrowing $\rightarrow$ new preregistration.

It follows Paper 69 [7], Paper 70 [8], and Paper 71 [9].

This paper does not modify the preregistered Paper-68 gate, does not reinterpret the Paper-69 outcome, does not define a new MAAT version, and does not execute a Gate+ L solver run. It freezes a new SAT-specific hypothesis for future independent execution.

This distinction matters. If occurrence/degree residuals are useful, that must be shown in a new predeclared experiment. They cannot be used retroactively to repair the first Gate Challenge result.

2 Background

The completed Gate Challenge Cycle I is archived as Research Series II:

<https://doi.org/10.5281/zenodo.21062386>.

The relevant cycle produced three facts.

1. **Paper 69.** The frozen MAAT v1.7 gate was executed on a public SATLIB family-balanced smoke subset. It was not confirmed as a practical CDCL improvement and lost clearly to MOMS.
2. **Paper 70.** The MOMS advantage was diagnosed as decision-local structural information, especially short-clause and occurrence-related pressure.
3. **Paper 71.** Most local channels were found to be reconstructible from $(H, B, S, V, R_{\text{rob}}, G_{\text{gate}})$. However, two weak residual candidates survived as hypothesis-generating signals: variable-occurrence coefficient of variation and degree coefficient of variation.

Paper 72 therefore narrows the next test to the smallest candidate not already absorbed by the existing supports: decision-local occurrence/degree residuals.

3 Hypothesis and Null

3.1 Primary Hypothesis

Decision-local occurrence/degree residuals provide additional SAT-specific decision information beyond the frozen MAAT v1.7 gate.

The phrase “additional” means that the signal must survive after residualizing against the frozen support basis:

$$\mathcal{S}_{69} = \{H, B, S, V, R_{\text{rob}}, G_{\text{gate}}\}.$$

3.2 Null Hypothesis

The residual local channels provide no robust improvement over frozen gate, score-with- R , or classical SAT heuristics.

The null is broad on purpose. It is not enough for Gate+ L to improve one comparison while failing against the relevant alternatives. Minimum support requires improvement over the frozen MAAT baselines. Strong SAT support additionally requires improvement over classical SAT heuristics.

4 Frozen Local Channel Definitions

Let F be a CNF formula over variables $v = 1, \dots, n$ and clauses $C = 1, \dots, m$.

4.1 Raw Occurrence Channel

Let o_v denote the number of clauses in which variable v occurs, counting either sign:

$$o_v = \#\{C : v \in C \text{ or } \neg v \in C\}.$$

The raw occurrence channel is fixed as

$$O_{\text{raw}}(F) = \frac{\text{std}_v(o_v)}{\text{mean}_v(o_v) + \varepsilon}. \quad (1)$$

4.2 Raw Degree Channel

Let d_x be the Paper-69 incidence-degree vector used in the support construction. The raw degree channel is fixed as

$$D_{\text{raw}}(F) = \frac{\text{std}_x(d_x)}{\text{mean}_x(d_x) + \varepsilon}. \quad (2)$$

This quantity is intentionally close to the Paper-69 B -support construction. For that reason the residual, not the raw value, is the object under test.

4.3 Calibration-Only Residualization

Residualization is performed only on calibration folds. Let

$$Z_{\mathcal{S}}(F) = z(H, B, S, V, R_{\text{rob}}, G_{\text{gate}})$$

denote calibration-standardized support coordinates. On the calibration subset only, fit

$$\hat{O}_{\text{raw}} = \alpha_0 + \alpha^\top Z_{\mathcal{S}}, \quad (3)$$

$$\hat{D}_{\text{raw}} = \beta_0 + \beta^\top Z_{\mathcal{S}}. \quad (4)$$

The fitted coefficients and calibration means/standard deviations are then frozen and applied to the test subset without refitting.

The residuals are

$$O_{\text{res}}(F) = O_{\text{raw}}(F) - \hat{O}_{\text{raw}}(F), \quad (5)$$

$$D_{\text{res}}(F) = D_{\text{raw}}(F) - \hat{D}_{\text{raw}}(F). \quad (6)$$

After residualization, O_{res} and D_{res} are standardized using calibration-fold means and standard deviations only. Those z-score statistics are frozen and then applied to the test fold.

4.4 Primary Combined Local Channel

The primary local occurrence channel for Paper 73 is frozen as

$$L_{\star}(F) = \frac{1}{2} [z(D_{\text{res}}(F)) - z(O_{\text{res}}(F))]. \quad (7)$$

The sign convention is fixed from Paper 71 before any Paper-73 execution. It must not be altered after observing Paper-73 results. The $z(\cdot)$ operations in Eq. (7) refer specifically to the post-residualization, calibration-fold standardization described above.

5 Frozen Gate+ L Combination

The base gate is the frozen Paper-69 gate G_{gate} . Before applying a logit transform, clip it to

$$G_{\text{gate}} \in [10^{-6}, 1 - 10^{-6}].$$

The SAT-specific Gate+ L activation is fixed as

$$G_{72}(F) = \sigma(\text{logit}(G_{\text{gate}}(F)) + L_{\star}(F)), \quad (8)$$

where $\sigma(x) = 1/(1 + e^{-x})$.

No additional weight is fitted:

$$\lambda_L = 1.$$

This choice is deliberately conservative. It avoids turning Paper 73 into a search over local-channel strengths. In particular, λ_L must not be increased, decreased, or selected by inspecting Paper-73 outcomes.

The solver policy scope is also frozen: Gate+ L may replace only the root-level gate activation used by the Paper-69 gate policy. All other solver rules, parsing rules, scoring rules, and baselines must remain unchanged.

6 Dataset and Split Rules

6.1 External Data Only

Paper 73 must use public external SATLIB/DIMACS CNF archives. Synthetic internal generators may be used only for debugging and must not count as primary evidence.

Raw CNF files must not be committed to the repository. Every instance used in the test must appear in a manifest with source information and SHA256 hash. Solver execution is allowed only after manifest validation passes.

6.2 Family-Balanced Primary Benchmark

The primary benchmark is family-balanced:

$$N_{\text{family}} = 10$$

validated CNFs per benchmark family where available. If a family contains fewer than 10 validated CNFs, all validated instances from that family may be used, and the shortfall must be reported.

6.3 Calibration/Test Split

For each family, split deterministically using seed

$$s = 72072.$$

The split is

$$20\% \text{ calibration, } \quad 80\% \text{ test.}$$

Calibration instances may be used only to fit residualization coefficients and calibration z-score statistics. They may not be used to choose signs, thresholds, budgets, or success criteria.

7 Budgets

The primary execution budget is fixed as:

Quantity	Frozen value
Conflict budget	5000 conflicts per policy-instance pair
Runtime budget	30 seconds per policy-instance pair
Bootstrap resamples	10000
Bootstrap seed	72072

If an implementation cannot complete the primary benchmark under these limits, the incomplete execution must be reported as incomplete. The budget may not be increased after seeing comparative outcomes.

8 Baselines and Nulls

The mandatory baselines are:

Class	Baseline
Classical SAT	MOMS
Classical SAT	Jeroslow–Wang (JW)
Classical SAT	VSIDS
MAAT baseline	score-with- R
MAAT baseline	frozen gate-v1.7
Local null	Gate+shuffled- L within family

The shuffled- L null must preserve the family labels and the marginal distribution of $L_\star$, but destroy the instance-level alignment between local residuals and solver decisions.

9 Primary Metric

Let $C_i(P)$ be the compute cost of policy P on instance i . For a candidate policy A and comparator B , define the paired utility delta as

$$\Delta_i(A, B) = C_i(B) - C_i(A). \quad (9)$$

Thus

$$\Delta_i(A, B) > 0$$

means that A used less compute than B .

The primary reported quantity is the mean paired delta over the test set:

$$\overline{\Delta}(A, B) = \frac{1}{N_{\text{test}}} \sum_i \Delta_i(A, B).$$

A 95% paired bootstrap confidence interval must be computed over test instances.

10 Success and Failure Criteria

10.1 Minimum Support

The Local Occurrence Hypothesis receives minimum support only if Gate+ L beats both frozen MAAT baselines:

$$\text{CI}_{95}^{\text{low}} [\overline{\Delta}(\text{Gate} + L, \text{frozen gate})] > 0,$$

and

$$\text{CI}_{95}^{\text{low}} [\overline{\Delta}(\text{Gate} + L, \text{score-with-R})] > 0.$$

10.2 Strong SAT Support

Strong SAT support requires minimum support and additionally:

$$\text{CI}_{95}^{\text{low}} [\overline{\Delta}(\text{Gate} + L, \text{MOMS})] > 0,$$

and

$$\text{CI}_{95}^{\text{low}} [\overline{\Delta}(\text{Gate} + L, \text{JW})] > 0.$$

10.3 Failure

The failure criteria are tiered, matching the two support levels above. This avoids treating failure against the strongest classical baselines as equivalent to failure against the frozen MAAT baselines.

No minimum support. The hypothesis receives no minimum support if Gate+ L fails to beat frozen gate or score-with- R with a positive 95% confidence lower bound, or if the apparent MAAT-baseline improvement disappears under the shuffled- L null.

No strong SAT support. The hypothesis receives no strong SAT support if it fails to beat MOMS or Jeroslow–Wang with a positive 95% confidence lower bound.

Failure is not a methodological problem. It would mean that the weak occurrence/degree residuals identified in Paper 71 do not provide robust SAT-specific decision value under external family-balanced evaluation.

10.4 Ambiguous Result: Minimum Support Only

If Gate+ L beats frozen gate and score-with- R but does not beat MOMS or Jeroslow–Wang, Paper 73 must report the result as *minimum support only*. This category is intentionally not a strong SAT result. It would mean that the occurrence/degree residuals improve the frozen MAAT baseline comparison but have not yet demonstrated practical competitiveness against classical SAT heuristics.

11 What Paper 72 Does Not Claim

This paper makes no claim that occurrence residuals are a new universal MAAT support. It makes no claim that Gate+ L will beat MOMS. It makes no claim that Paper 69 was incomplete or incorrectly evaluated. It does not introduce MAAT v1.8.

The correct interpretation is narrower:

Paper 72 freezes the smallest SAT-specific residual hypothesis that survived the Paper-69–71 diagnostic cycle, so that it can be fairly tested later.

12 Reproducibility

The protocol artifact is located at:

`experiments/local_occurrence_hypothesis_paper72/`

Running

```
python3 local_occurrence_protocol_paper72.py
```

writes:

`outputs_paper72_local_occurrence_protocol/paper72_protocol.json`

The script writes a machine-readable preregistration file only. It does not load CNFs, does not compute features, does not calibrate residualization models, and does not execute any solver policy.

13 Conclusion

Paper 72 marks the transition from post-hoc diagnosis to a new predeclared SAT-specific test. The completed Gate Challenge Cycle I showed that the frozen v1.7 gate was not supported in SAT, that MOMS uses decision-local structural information, and that most such information is already reconstructible from existing MAAT supports. Only weak occurrence/degree residuals remain as plausible candidates.

The Local Occurrence Hypothesis is therefore intentionally narrow. It will be supported only if the frozen Gate+ L protocol improves practical CDCL utility over frozen gate and score-with- R , and strongly supported only if it additionally defeats MOMS and Jeroslow–Wang under family-balanced external evaluation. If it fails, the result should be reported as a clean negative outcome.

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